Formulas

t_e = expected time

t_p = pessimistic time

t_o = optimistic time

t_m = most likely

S_t = Standered deviation

V_t = Varience

PERT

*For Finding expected tim

$$t_e = t_o + 4t_m + t_p / 6$$

*For finding standered diviation:

$$S_t = t_p - t_o / 6$$

SD = $\sqrt{\text{Variance}}$

*For finding variance

$$v_t = (t_p - t_o / 6)^2$$

el

duration & activities

Pert

Item cost = C_1

Purchasing/setup/ordering/Acquisition = C_2

Inventory/holding = C_3

Shortage cost = C_4

E-O-Q (Economic order quantity)

$Q^* = \sqrt{2C_2D/C_3} \Rightarrow Q^*$ is E-O-Q

$Q^*/D \Rightarrow t^*$ is time period

$D/Q^* \Rightarrow N^*$ is Number of Orde

For optimum order Quantity

$$Q^* = \sqrt{\frac{2C_2D}{C_3}} \times \sqrt{\frac{C_3 + C_4}{C_4}}$$

For Shortage optimum

$$S^* = \sqrt{\frac{2C_2D}{C_4} \times \frac{C_3}{C_3 + C_4}}$$

Maximum inventry

$$I = Q^* - S^*$$

Total cost Per Period

$$C_1 Q^* + C_2 + \frac{C_3 (Q^* - S^*)}{2D} + \frac{C_4 S^{*2}}{2D}$$

time of manufacturing

$$t_1 = \frac{Q^*}{R}$$

optimum manufacturing Quantit

$$Q^* = \sqrt{\frac{2C_2 D}{C_3}} \times \sqrt{\frac{R}{R-D}}$$

3

Optimum annual cost

Item cost + ordering cost + Holding cost

Types and formulla's of floats

* Indepent float

$I = J_E - i_L - D$

* Free float

$F = J_E - i_E - D$

* Total float

$I = J_L - i_E - D$

* Earliest start time (EST)
* Latest finish time (LFT)
* Earliest finish time (EFT)

EFT = EST + Duration of activity.

* Latest start time (LST)

LST = LFT - Duration of activity.

Best of luck :-)